

1928

K. N. Toosi University
of Technology



Faculty of Mechanical Engineering

Hybrid Vehicles Project Report

Conceptual Design and CAD Modeling of a Formula E Powertrain System

By

Amirhossein Mohammadi

Supervisor

Dr. Amir Taghavipour

June 2025

Abstract

This report examines the powertrain system in Formula E cars, focusing on its key components battery, inverter, electric motor, transmission and their integration for optimal performance. The study highlights design considerations, such as acceleration performance, torque-speed characteristics, and single-speed transmission optimization. Mathematical models are used to analyze power requirements, top speed, and acceleration for conceptual design and components sizing. Additionally, it discusses how Solidworks as the CAD software is employed to validate component geometries and simple mechanical integration.

Keywords: Formula E, Powertrain, Electric Motor, Inverter, Transmission, Conceptual Design, Computer-Aided-Design

Table of Contents

1. Introduction 1

 1.1. History and Background 1

 1.2. Scope and Project Definition 2

 1.3. Electric Motor 2

 1.4. Inverter 4

 1.5. Transmission 5

2. Conceptual Design and Component Sizing 7

 2.1. Transmission Calculation 7

 2.2. Traction Motor Calculation 8

 2.3. Inverter Calculations 8

 2.4. Summary of Results 9

3. CAD Implementation 9

Conclusion 11

References 12

1. Introduction

1.1. History and Background

Formula E, sanctioned by the Fédération Internationale de l'Automobile (FIA), was established in 2014 as the world's first all-electric international motorsport championship. Conceived to promote sustainable racing and accelerate EV technology, the series has become a global platform for automotive innovation, blending competition with environmental advocacy.

The championship's evolution is marked by distinct generations of race cars. The inaugural Gen1 era (2014–2018) introduced electric racing to mainstream audiences, featuring futuristic designs and street-circuit races in major cities. With Gen2 (2018–2022), Formula E matured into a high-performance series, adopting aggressive aerodynamics and longer battery ranges. The current Gen3 cars (2023–present) represent the pinnacle of efficiency, with lighter designs, faster charging, and enhanced regenerative braking systems.

Under the FIA's regulatory framework, Formula E balances cutting-edge engineering with cost control, allowing manufacturers to experiment within defined technical boundaries. The series has attracted major automakers, from Audi and Jaguar to Porsche and McLaren, all leveraging the championship to advance electric mobility.



Figure 1.1. NEOM McLaren Gen3 at the Berlin ePrix

1.2. Scope and Project Definition

Formula E represents the pinnacle of electric motorsport, where powertrain efficiency is critical for performance. Unlike internal combustion engines, Formula E cars rely on a fully electric powertrain, comprising battery as the energy storage, Inverter for DC to AC conversion, electric motor as the traction unit and transmission for transferring power and torque-speed modification. This report examines the conceptual design of a Formula E powertrain and integrating theoretical analysis with CAD modeling. It should be noted that the battery is not analyzed in detail within this study, as it falls outside the defined scope of the powertrain subsystem. The conceptual design of the battery is instead being developed by another teammate independently as part of the vehicle's broader energy storage system. Figure 1.2 shows different main components of the powertrain system in a Formula E car.

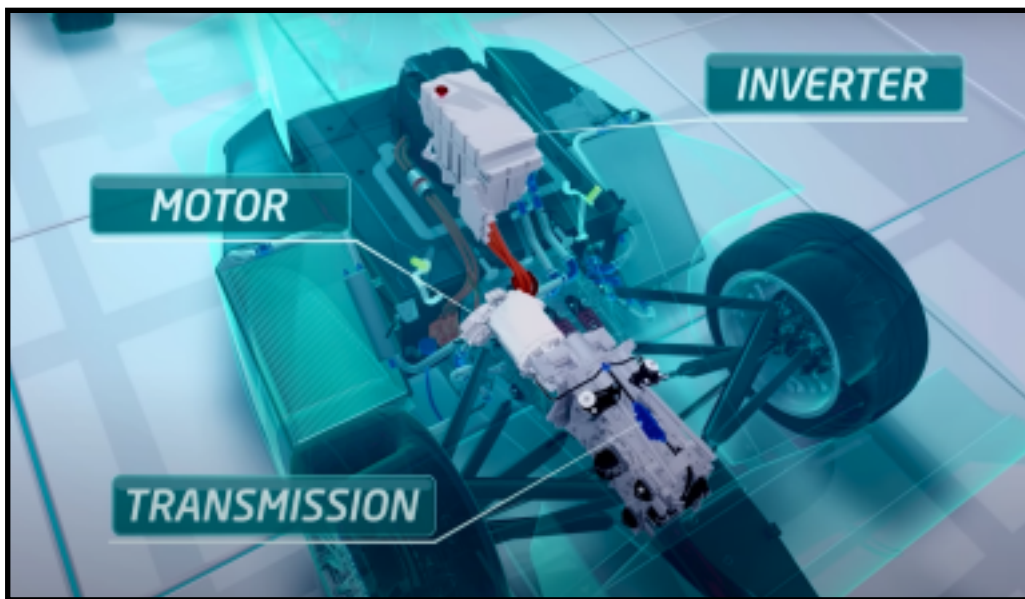


Figure 1.2. Main Components of the Powertrain System

1.3. Electric Motor

Modern Formula E Gen3 cars utilize advanced permanent magnet synchronous motors that represent a significant evolution from

conventional brushed DC motors historically used in automotive applications.

The contemporary design reverses the traditional architecture by employing a rotating permanent magnet rotor surrounded by stationary copper windings, eliminating the need for inefficient carbon brushes while improving thermal management capabilities as shown in Figure 1.3.



Figure 1.3. Rotor and Stator of the PMSM

These brushless motors operate on three-phase AC power converted from the 800V DC battery system through sophisticated silicon carbide inverters, enabling precise control of electromagnetic fields. A key characteristic of Gen3 powertrains is their optimization for high-RPM operation, typically exceeding 20,000 revolutions per minute, which prioritizes energy efficiency over maximum torque output in accordance with the series' strict energy management requirements.

The compact packaging of these motors, with typical dimensions about 40 centimeters in length and 30 centimeters in diameter, demonstrates remarkable power density while maintaining structural integrity against substantial centrifugal forces. Advanced cooling systems with integrated liquid cooling jackets maintain optimal operating temperatures for both the stationary windings and power electronics. This design philosophy reflects Formula E's role in pushing the boundaries of electric motor technology, where the constraints of energy-limited competition have driven innovative solutions that balance performance, efficiency, and reliability under extreme operating conditions. The transition from basic automotive starter motor concepts to these highly

specialized racing units illustrates the rapid advancement of electric propulsion systems fostered by the championship's unique technical challenges. Figure 1.4 shows the outer part of the permanent magnet synchronous motor which can be an inspiration for CAD design.

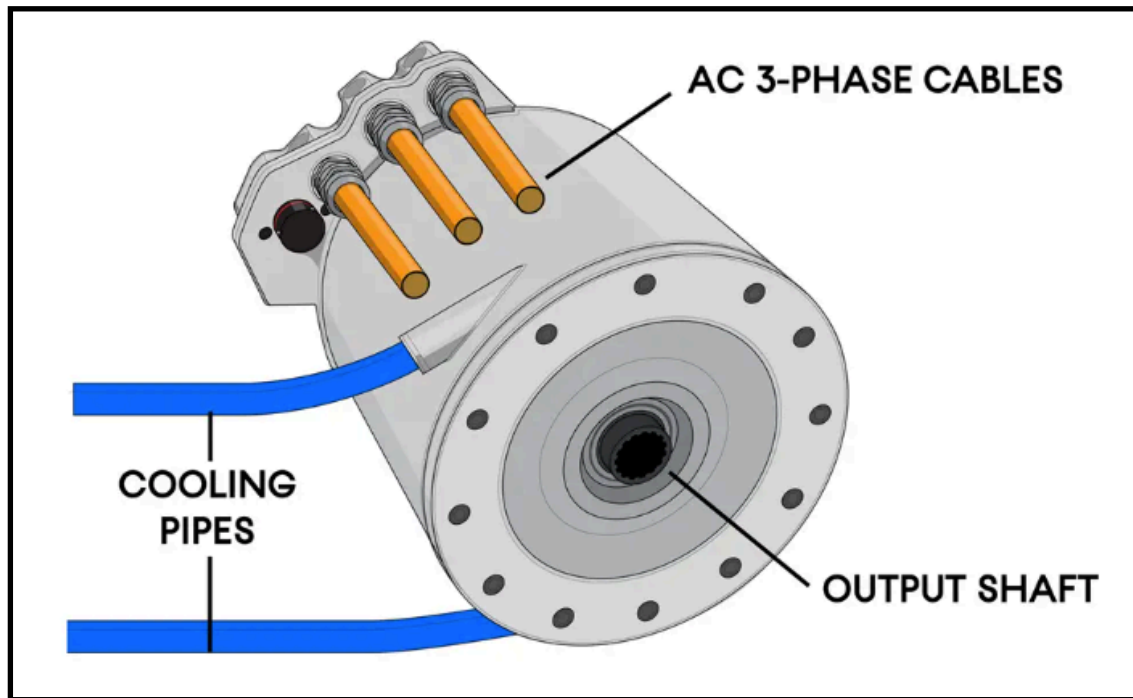


Figure 1.4. Permanent Magnet Synchronous Motor

1.4. Inverter

It's the job of the inverter to do all the electrical conversion from the DC of the battery to the high frequency high voltage AC for the Motor. This inverter/rectifier electrical function can be done simply with diodes, but for the level of performance required for an electrical vehicle, this requires something more complex, so the switching is achieved with semiconductor technology at the high frequency of 10000 pulses per second (Hz).

At its core the inverter has a power circuit-board, the two DC (battery) cables (single-phase) come in one side and the three AC (motor) cables (three-phase) the other. They pass via bus bars to these switches, which are controlled by a separate logic circuit-board, in order to translate

the current from one form to another. The quicker the switches can switch, the more power they can handle and fewer electrical losses along the way the better the inverter's performance will be.

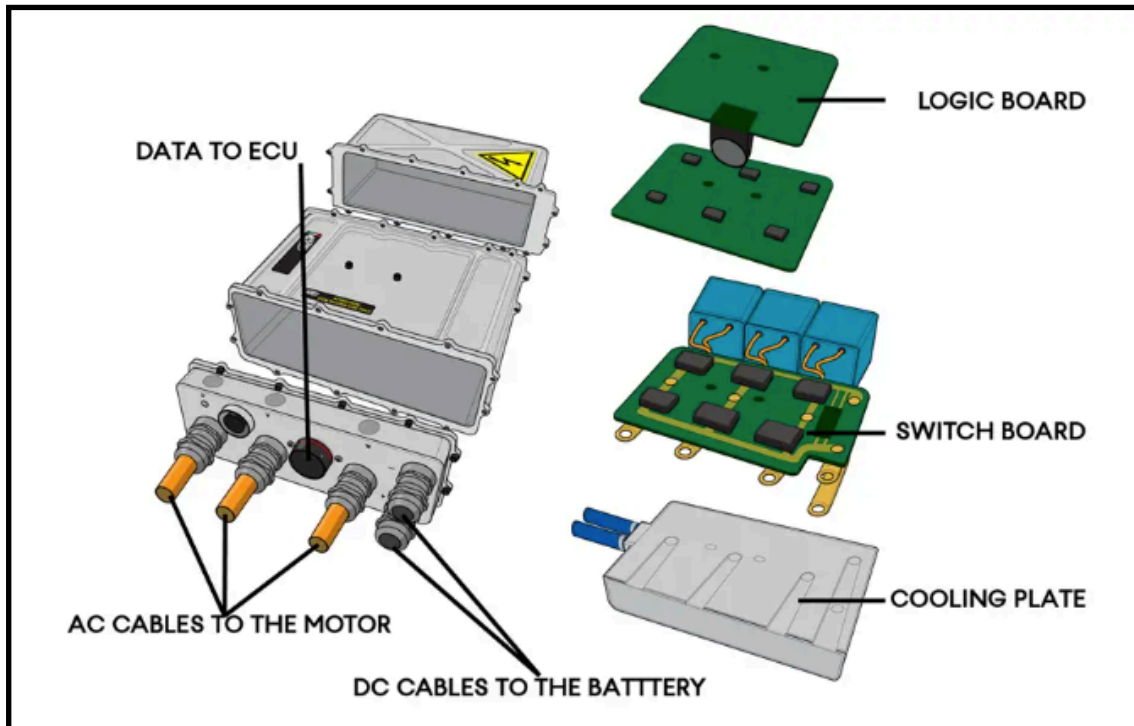


Figure 1.5. Inverter

1.5. Transmission

Formula E cars utilize a single-speed transmission, a design choice driven by the unique characteristics of electric motors and the series' technical requirements. Unlike internal combustion engines, which require multiple gears to operate efficiently across a wide speed range, electric motors deliver near-instant torque and maintain high efficiency over a broad RPM band. This eliminates the need for complex multi-speed gearboxes while reducing weight, mechanical losses, and maintenance requirements.

The single-speed transmission is optimized to balance acceleration and top speed within Formula E's energy constraints. Electric motors can achieve extremely high rotational speeds (up to 20,000 RPM in Gen3 cars),

allowing a fixed gear ratio to efficiently cover the full speed range of the race. Additionally, the simplicity of a single-speed system enhances reliability.

By forgoing gear shifts, Formula E powertrains maximize energy efficiency, reduce drivetrain complexity, and align with the series' focus on sustainable, high-performance electric racing. This approach demonstrates how electric motors enable innovative engineering solutions that differ fundamentally from traditional motorsport designs. Figure 1.6 shows a single-speed / single-step gearbox including helical bevel and spur gears that can be inspiration for cad design due to simplicity while Figure 1.7 demonstrates a more complex single-speed gearbox with multi-step transmission.

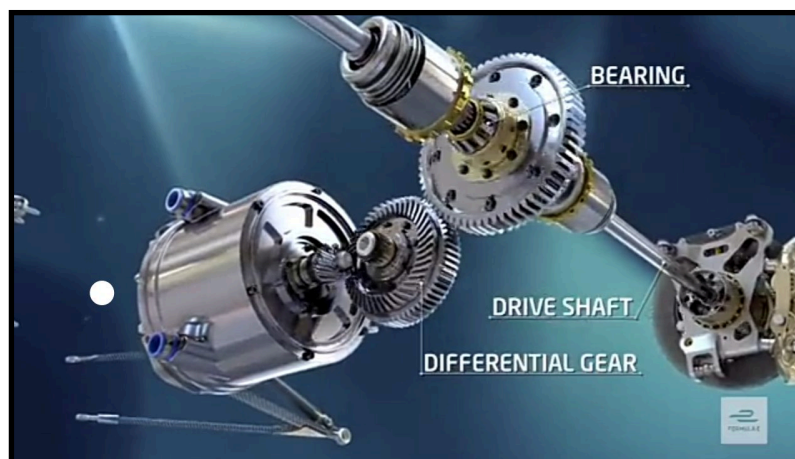


Figure 1.6. Single-Stage Transmission

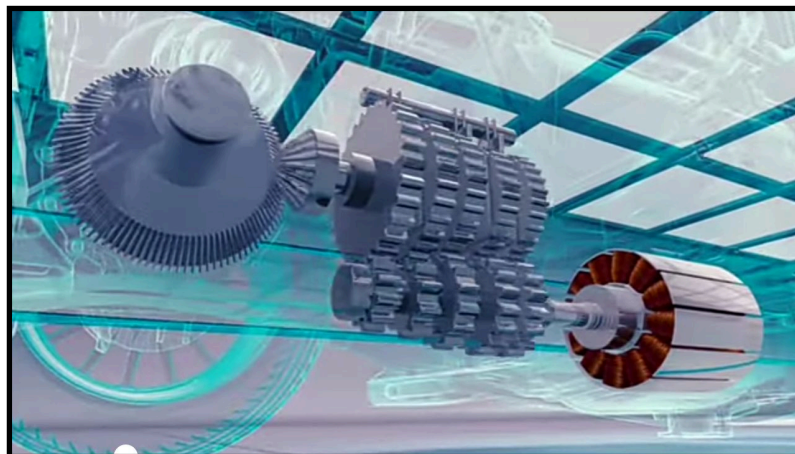


Figure 1.7. Multi-Stage Transmission

2. Conceptual Design and Component Sizing

In this section, each of the three components of transmission, motor and inverter are designed based on design parameters and performance specifications (criteria) which is based on longitudinal acceleration performance. The required information is expressed in Table 2.1 and 2.2 based on Gen3 cars and some of the others are logically assumed.

Table 2.1. Design Parameters

Vehicle Total Mass (kg)	760/840
Rolling Resistance Coefficient	0.008 - 0.012
Aerodynamic Drag Coefficient	0.28 - 0.32
Front Area (m ²)	1.4 - 1.6
Transmission Efficiency (%)	95
Tyre Diameter (cm)	68
Maximum Rotational Speed of the Motor (rpm)	20000

Table 2.2. Performance Specifications (Design Criteria)

Acceleration Time (s)	1.8 - 2.5
Top Speed / Cruising Speed(km/h)	322

2.1. Transmission Calculations

Assuming no slip condition and maximum longitudinal velocity at the maximum rotational speed of the motor, the gear or speed ratio of the transmission can be calculated as follows.

$$V_{max} = R_{Wheel} \omega_{Wheel} \text{ (Equation 2.1)}$$

$$\omega_{Wheel} = \frac{\eta_{tf} \omega_{Motor,max}}{N_{tf}} \frac{2\pi}{60} \text{ (Equation 2.2)}$$

Combining Equation 2.1 and 2.2, leads to the following equation that is used to obtain speed ratio.

$$N_{tf} = \frac{\pi R_{Wheel} \omega_{Wheel} \eta_{tf}}{30 V_{max}} \Rightarrow N_{tf} = 7.56$$

2.2. Traction Motor (EM) Calculations

From vehicle dynamics and acceleration performance equations based on laws of motion, the maximum tractive power can be obtained. Equation 2.3 represents the corresponding power which is the design criterion of the motor and can be considered as its rated power. Note that the base speed is assumed to be one quarter of the maximum speed.

$$\delta = 1 + 0.04 + 0.0025 N_{tf}^2$$

$$P_t = \frac{\delta M}{2 t_a} (V_f^2 + V_b^2) + \frac{2}{3} M g f_r V_f + \frac{1}{5} \rho_a C_d A_f V_f^3 \text{ (Equation 2.3)}$$

$$\Rightarrow P_t \approx 320(kW)$$

2.3. Inverter Calculations

Inverter is sized based on its input DC current and output AC current. These values are calculated from the following equations.

$$I_{Inverter, Input} = I_{DC} = \frac{P_{PPS}}{\eta_{Inverter} V_{DC}} = \frac{P_{Motor}}{\eta_{Motor} \eta_{Inverter} V_{DC}} \text{ (Equation 2.4)}$$

$$I_{Inverter, Output} = I_{AC} = \frac{P_{Motor}}{\eta_{Motor} V_{AC} PF \sqrt{3}} \text{ (Equation 2.5)}$$

$$\Rightarrow I_{DC} \approx 500(A), I_{AC} \approx 385(A)$$

2.4. Summary of Results

Sizing specifications of the components are shown in Table 2.3. By having these specifications, component selection is possible.

Table 2.3. Summary of Results

Traction Motor (EM)	Transmission	Inverter
Power = 320 - 350(kW)	Gear (Speed) Ratio = 7.56	Input Current = 500(A)
Rotational Speed = 20000(rpm)	Input Power = 350(kW)	Output Current = 385(A)
Torque = 150-167(N.m)	Input Rotational Speed = 20000(rpm)	Frequency = 10000(Hz)

3. CAD Implementation

In the final chapter of this report, powertrain system components are design using Solidworks. Since the purpose of the project is the concepts as mentioned in the previous chapters, the focus is on simplified CAD models. Therefore, these components are not designed in details. The dimensions are mostly based on reverse engineering of predesigned models and the mentioned numbers in introduction and some less important of them such as input cables or pipes are logically assumed because of being out of context. Figures 3.1-3.4 shows the CAD model of reviewed components. All of the associated parts and assemblies are attached to this report.

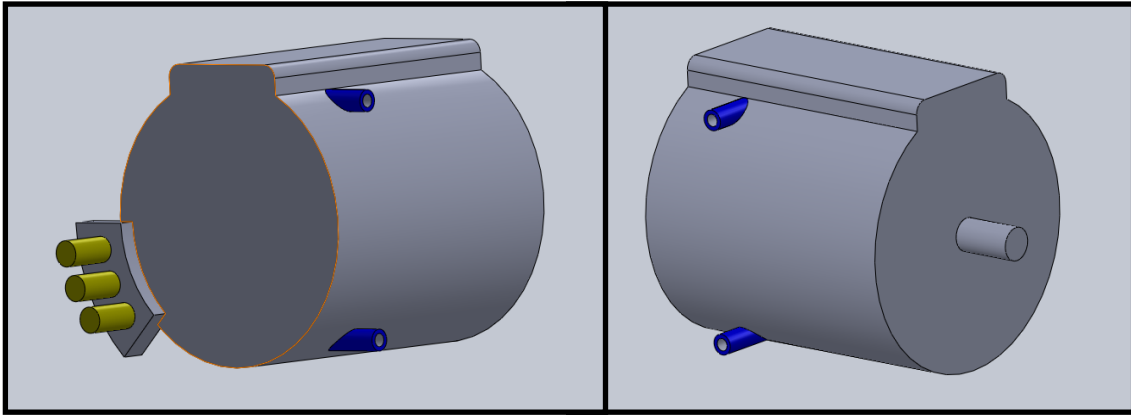


Figure 3.1. Electric Motor CAD File

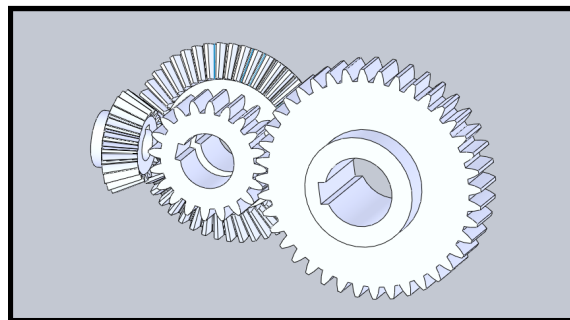


Figure 3.2. Transmission CAD File

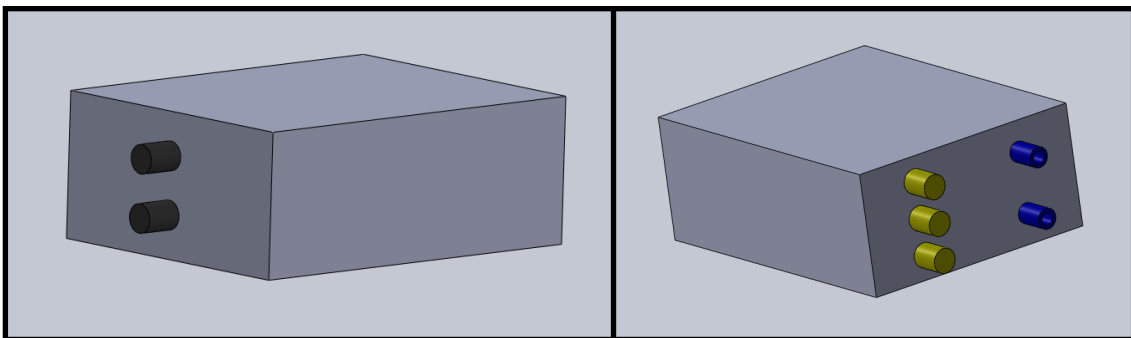


Figure 3.3. Inverter CAD File

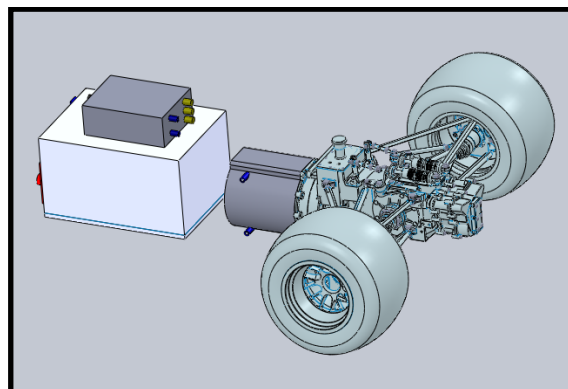


Figure 3.4. Powertrain Assembly File

Conclusion

This study presented a conceptual design methodology for Formula E powertrain systems, combining analytical modeling with CAD-assisted validation. Mathematical analysis of power requirements, acceleration performance and single-speed transmission optimization established key performance benchmarks, while Solidworks provided critical spatial verification of the proposed architecture.

As a conceptual study, this work prioritizes foundational sizing principles over detailed manufacturing specifications, establishing a framework for subsequent development phases.

This methodology provides motorsport engineers with a replicable process for initial powertrain conceptualization, where rapid validation of design assumptions is as critical as performance calculations. Future work should transition from conceptual to detailed design, incorporating advanced multi-physics simulations.

References

[1] www.wikipedia.org

[2] www.carmagazine.co.uk

[3] youtube.com

[3] Amir Taghavipour, “Principles of Hybrid Powertrain Systems Design”, Lecture, K.N.Toosi University of Technology, 2025.

[4] Thomas D. Gillespie, “Fundamentals of Vehicle Dynamics ”, Society of Automotive Engineers, 1994.